

Kivan Polimis

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Senior Data Scientist & Applied Machine Learning Engineer with over 13 years of experience architecting systems to model and predict behavioral patterns and population dynamics. Proven track record of delivering measurable business impact, including \$100M in projected annual savings. Expert in Python/R and behavioral modeling, currently specializing in production ML pipelines and containerized microservices (Docker).

EMPLOYMENT

2021–Present Senior Data Scientist, Karna, Atlanta, GA 30345

- Architected and deployed a web-based document intelligence platform integrating cloud document extraction services (Azure, AWS, GCP) and GPT-4 for automated clinical document processing; led client demos with government health agencies; primary showcase for document AI.
- Developed stakeholder dashboards and led client-facing analytics delivery for a live state Medicaid engagement, coordinating across three organizations and presenting MVP to state agency and health plan audiences.
- Designed a production ML pipeline for patient risk classification using XGBoost and MLflow with automated drift monitoring, reducing unnecessary ED visits and downstream costs among high-risk patients.
- Spearheaded the design of robust, compliant data infrastructure using R, Python, and shell scripting within Snowflake to create reproducible statistical analyses for national opioid response planning.
- Architected a HIPAA/NIST-compliant 3-tier AWS environment and a low-latency Flask API to ensure secure data access while reducing regulatory compliance risk.

Awards: President's Award (2023), Technical Achievement Award (2023), Community Engagement Award (2022), Outstanding Publication runner-up (2022), Technical Achievement runner-up (2024)

2021–Present Lecturer, University of Washington, Department of Sociology, Seattle, WA 98195

- Developed curricula on the technical and societal impacts of algorithmic decision-making, focusing on model interpretability, algorithmic fairness metrics, and mitigating statistical bias in automated systems.
- Instructed data science labs utilizing Python and R for the ingestion, feature extraction, and analysis of digital trace data and unstructured datasets.

2020–2021 Senior Policy Advisor, Office of Harris County Judge Lina Hidalgo, Houston, TX 77002

- Optimized mission-critical ETL pipelines for COVID-19 reporting, slashing processing latency by 90% and strengthening data quality for real-time decision making by policymakers.
- Executed county-wide initiatives by collaborating intra-government and with private sector; estimated \$6 million annual savings from permanent supportive housing proposals.

2020–Present Principal and Founder, Atlas Analytics, Houston, TX 77003

- Built a scalable entity resolution pipeline in Python/SQL to join fragmented SBA datasets with merchant registries, achieving an 86% recall using a tiered probabilistic matching algorithm to disambiguate complex corporate linkages.
- Applied NLP and advanced statistical modeling to user-generated text data, optimizing the marketing funnel and increasing referral conversion by 20%.
- Architected end-to-end ML pipelines from feature engineering through model deployment, delivering customer propensity scoring and market segmentation solutions.

2018–2020 Data Scientist, MAANA, Houston, TX 77002

- Engineered and deployed ML micro-services using Docker and Kubernetes for real-time vessel allocation optimization, reducing route planning from hours to minutes.
- Conducted cost baseline forecast analysis of historical trades projecting \$100 million in annual savings.

2017–2018 Postdoctoral Researcher, Università Bocconi, Milan, Italy 20136

Carlo F. Dondeña Centre for Research on Social Dynamics and Public Policy

- Engineered multi-source data fusion pipelines utilizing unstructured digital trace data and social media APIs to nowcast migrant stocks with high-frequency temporal resolution.
- Developed predictive frameworks to evaluate public policy outcomes, applying advanced demographic methods and longitudinal analysis to quantify the impact of environmental and political shocks.

APPLIED MLE PROJECTS

Multi-Agent Clinical Documentation Platform

- Designed a modular microservices architecture with a discoverable agent registry for LLM-powered clinical documentation and ICD-10 coding automation; implemented versioning, observability, and containerized deployment via PyTorch/TensorFlow.
- Built a Majority-Vote Ensemble classifier for high-dimensional financial transaction data (TF-IDF/XGBoost vs. Sentence-BERT), yielding a 10% accuracy improvement over baseline.

Transaction Text Classifier

- Developed a transaction text classification pipeline for personal finance data, evaluating TF-IDF + Logistic Regression, LinearSVC, and XGBoost against Sentence-BERT transformer architectures.
- Implemented a Majority-Vote Ensemble achieving 89.7% accuracy, with classical TF-IDF models matching transformer-based performance.

TECHNICAL SKILLS

Instructor	Software Carpentry Instructor, Data Carpentry Instructor
Cloud/DevOps	AWS (ECR, SageMaker, Lambda, S3), Azure (ACR, AKS, Azure Functions), Docker, Git, CI/CD
Programming	Python, SQL, R, shell scripting
ML & Analytics	XGBoost, LightGBM, scikit-learn, PyTorch, TensorFlow, MLflow, NLP/text-as-data methods, FastAPI
Project Tools	Jira, GitHub Projects
Open Source	forest-confidence-interval (scikit-learn-contrib)
Certifications	Software Carpentry Instructor, Data Carpentry Instructor, AWS Machine Learning Engineer -- Associate (In Progress), AWS Solutions Architect -- Associate (In Progress)

EDUCATION

- 2017 Ph.D.**, Computational Social Science, University of Washington
Center for Statistics and the Social Sciences: Certification in Social Statistics
- 2012 M.A.**, Sociology, University of North Carolina at Chapel Hill
Carolina Population Center Pre-doctoral Fellow
- 2009 B.A.**, Sociology, Princeton University

SELECT PUBLICATIONS

- 2020** Alexander, Monica, **Kivan Polimis**, and Emilio Zagheni. “Combining social media and survey data to nowcast migrant stocks in the United States”. *Population Research and Policy Review*.
- 2020** Salganik, Matthew, Ian Lundberg, Alexander T. Kindel, Caitlin E. Ahearn, et al.. “Measuring the predictability of life outcomes with a scientific mass collaboration”. *Proceedings of the National Academy of Sciences*.
- 2019** Alexander, Monica, **Kivan Polimis**, and Emilio Zagheni. “The impact of Hurricane Maria on out-migration from Puerto Rico: Evidence from Facebook”. *Population and Development Review*.

VOLUNTEER EXPERIENCE

- 2023–Present** Communications Vice Chair, **Princeton Alumni Association of Houston**